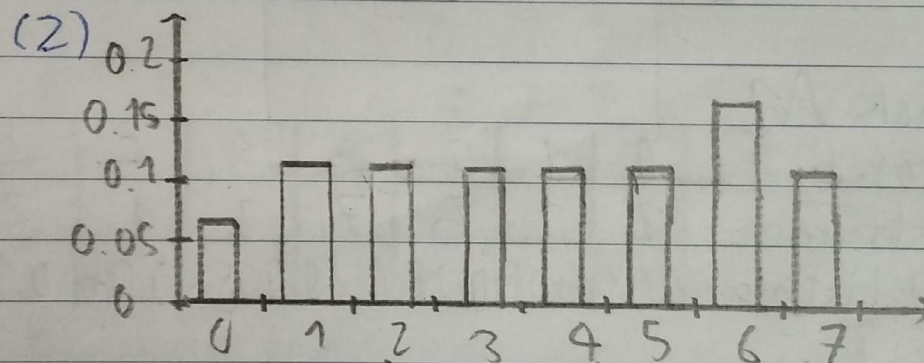
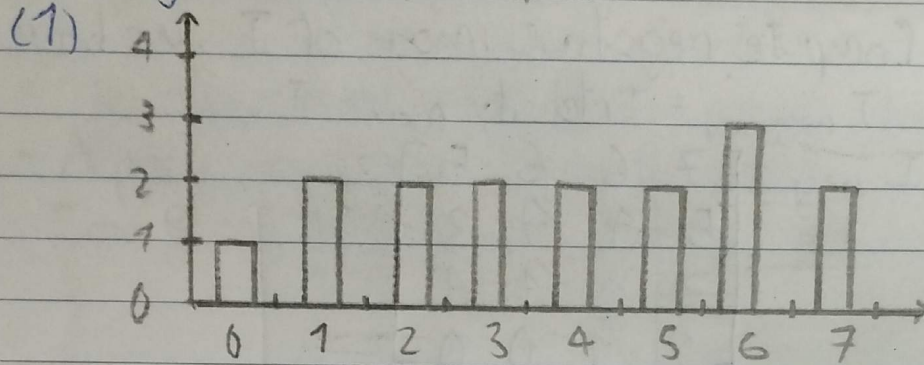


Labwork 1: Histogram and Point Processing

Part A

Task A1

Intensity	0	1	2	3	4	5	6	7
(1) Histogram	1	2	2	2	2	2	3	2
(2) Normalized Histogram	0.0625	0.125	0.125	0.125	0.125	0.125	0.1875	0.125



Equalized histogram

Intensity	0	1	2	3	4	5	6	7
cdf	1	3	5	7	9	11	14	16
N.cdf	0.0625	0.1875	0.3125	0.4375	0.5625	0.6875	0.8125	1
N.cdf.L	0.4375	1.3125	2.1875	3.0625	3.9375	4.8125	5.6875	6.5625
New intensity	0	1	2	3	4	5	6	7
$\Rightarrow M = I =$	$\begin{bmatrix} 0 & 1 & 1 & 2 \\ 2 & 3 & 3 & 4 \\ 4 & 5 & 6 & 6 \\ 5 & 6 & 7 & 7 \end{bmatrix} \Rightarrow \text{same histogram as (1)}$							

The gray-level value of all pixels in image I is

[0 1 1 2 2 3 3 4 4 5 5 6 6 6 7 7]

$\Rightarrow k = 4$

Thresholding

$$I \rightarrow B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 7 & 7 & 7 \\ 7 & 7 & 7 & 7 \end{bmatrix} \quad \left(\begin{array}{l} \text{if } I_{x,y} > k \Rightarrow 7 \\ I_{x,y} \leq k \Rightarrow 0 \end{array} \right)$$

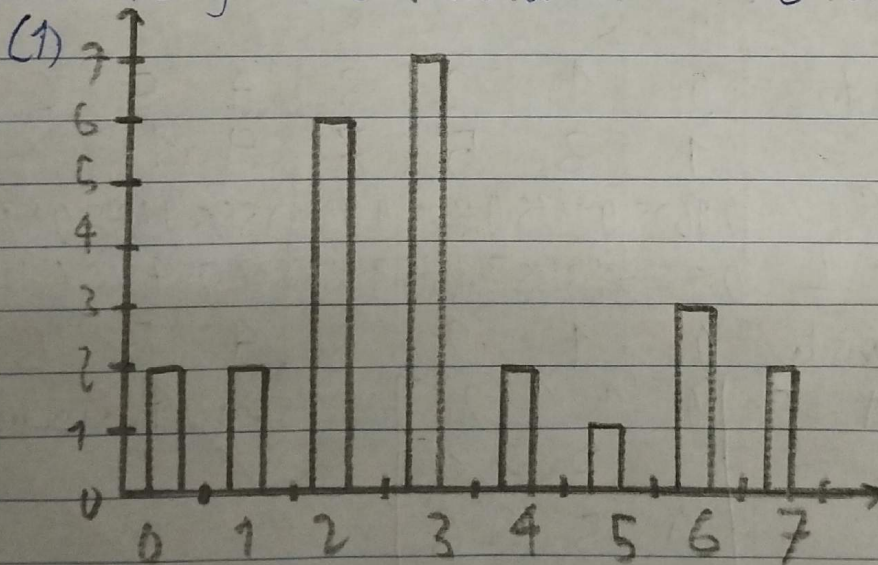
To compute negative image of I , we have

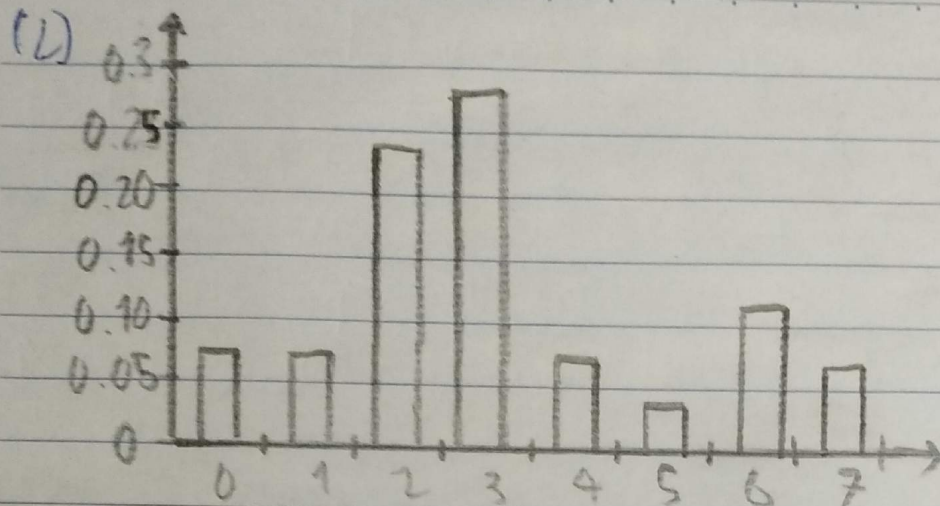
$$I_{neg\ x,y} = \text{Intensity max} - I_{x,y}$$

$$\Rightarrow I_{neg} = \begin{bmatrix} 7 & 6 & 6 & 5 \\ 5 & 4 & 4 & 3 \\ 3 & 2 & 1 & 1 \\ 2 & 1 & 0 & 0 \end{bmatrix}$$

Task A2:

Intensity	0	1	2	3	4	5	6	7
(1) Histogram	2	2	6	7	2	1	3	2
(2) N. histogram	0.08	0.08	0.24	0.28	0.08	0.04	0.12	0.08





- The gray-level value that appears the most, is 3
- Applying threshold with $K=3$

$$I \rightarrow B = \begin{bmatrix} 0 & 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 7 & 7 \\ 0 & 0 & 0 & 0 & 7 \\ 0 & 0 & 7 & 7 & 7 \end{bmatrix}$$

- There are 8 pixels assigned to the foreground and 17 pixels to the background
- The selected threshold $K=3$ separate well because there are much more low-intensity pixels ≤ 3 than high-intensity pixels in the image